

KOELN BONN, Germany

WMO: 105130

Lat: 50.8644N

Lon: 7.1575E

Elev: 326

StdP: 14.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	21.6	8.7	8.7	23.3	14.0	11.3	25.3	26.2	45.6	22.9	45.6	4.6	100	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	87.9	67.6	84.0	66.0	80.4	64.8	69.8	82.6	68.1	79.5	66.6	77.0	8.3	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	97.0	73.7	64.3	91.4	72.3	62.7	86.2	70.5	34.0	82.7	32.7	79.2	31.3	77.2	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.6	15.7	DB	10.9	94.6	6.3	2.7	6.3	96.6	2.6	98.2	-0.9	99.7	-5.5	101.7	(4)
(5)			WB	10.1	73.3	6.2	1.6	5.6	74.5	2.0	75.4	-1.5	76.4	-5.9	77.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	37.0	38.4	43.9	50.9	56.9	62.9	66.9	65.2	59.3	52.5	44.4	39.1
	DBStd	12.44	7.76	7.65	7.19	7.23	6.93	6.36	6.28	5.66	5.45	6.75	6.97	7.72
	HDD50	1621	404	326	212	75	16	0	0	0	1	52	192	343
	HDD65	5188	867	745	655	425	266	114	53	66	188	388	618	803
	CDD50	2178	2	2	21	102	229	388	524	472	279	130	24	5
	CDD65	271	0	0	0	2	14	52	112	74	16	1	0	0
	CDH74	2692	0	0	0	56	171	540	1063	718	135	9	0	0
	CDH80	831	0	0	0	4	34	154	379	233	27	0	0	0
Wind	WSAvg	7.3	8.5	8.0	8.0	7.2	7.1	6.8	6.6	6.3	6.4	7.1	7.7	8.0
Precipitation	PrecAvg	33.5	2.6	2.4	2.2	2.0	2.6	3.5	3.4	3.4	2.7	2.7	2.9	3.0
	PrecMax	39.6	5.2	5.8	4.2	3.8	4.9	7.2	7.4	7.5	6.5	6.5	5.2	6.1
	PrecMin	27.8	0.4	0.5	0.4	0.0	1.0	1.1	1.5	1.2	0.7	1.3	0.0	0.8
	PrecStd	3.7	1.3	1.2	1.1	0.9	1.1	1.3	1.4	1.8	1.5	1.2	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.8	59.4	69.4	80.2	84.4	89.7	93.6	91.7	84.2	75.0	62.8	57.3
		MCWB	51.4	47.5	52.6	58.1	65.3	69.9	68.9	68.8	66.4	62.9	55.9	52.0
	2%	DB	53.3	53.9	63.0	74.9	78.5	84.3	89.1	86.2	77.4	69.0	59.2	54.6
		MCWB	48.9	47.5	50.6	57.1	61.8	66.2	68.0	66.9	64.3	60.3	54.1	50.5
	5%	DB	50.0	51.6	59.1	69.6	74.8	80.4	84.2	81.5	73.3	65.0	57.0	52.1
		MCWB	46.8	46.6	49.5	55.1	60.0	65.1	66.1	65.5	62.7	58.5	52.5	48.3
	10%	DB	47.3	48.8	55.3	64.4	70.6	76.1	80.3	77.3	69.5	62.4	53.9	49.8
		MCWB	44.4	45.0	47.9	52.8	58.4	63.4	65.3	63.9	60.6	57.1	50.4	46.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	51.0	55.5	60.4	67.9	71.6	72.0	71.5	68.4	63.2	57.5	53.5
		MCDB	55.8	55.6	63.6	74.2	79.2	84.8	86.1	85.8	79.9	70.6	61.6	56.1
	2%	WB	49.5	48.7	52.6	57.8	64.6	68.7	69.9	69.0	65.5	61.3	54.8	51.0
		MCDB	52.4	52.5	60.1	71.1	74.9	80.1	82.7	81.5	75.2	67.4	58.4	54.2
	5%	WB	46.8	47.1	50.7	55.8	61.6	66.7	68.3	67.0	63.6	59.5	52.9	49.0
		MCDB	49.6	50.9	57.2	66.7	71.1	77.1	79.9	77.7	71.4	64.5	56.4	52.0
	10%	WB	44.3	45.1	48.8	54.0	59.3	64.5	66.7	65.4	61.7	57.6	50.8	46.6
		MCDB	46.8	48.7	54.8	63.6	68.1	73.6	77.2	74.8	68.1	61.8	53.9	49.3
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.0	15.4	19.0	18.6	18.9	19.1	18.9	17.1	14.2	9.8	8.8
		MCDBR	11.2	14.2	22.4	26.9	25.7	26.9	26.9	27.2	23.4	17.8	11.4	10.2
		MCWBR	9.1	9.3	12.4	12.8	12.1	12.3	10.3	11.5	11.9	10.5	8.0	8.0
	5% WB	MCDBR	10.7	13.0	18.5	23.3	21.8	22.4	22.9	22.3	19.9	15.8	10.2	9.8
MCWBR		9.1	9.3	10.8	11.6	11.3	11.4	9.7	10.7	10.7	9.9	7.9	8.4	
Clear-Sky Solar Irradiance	taub	0.317	0.343	0.374	0.410	0.415	0.418	0.423	0.416	0.386	0.362	0.338	0.315	
	taud	2.420	2.371	2.299	2.235	2.255	2.267	2.251	2.296	2.361	2.422	2.419	2.402	
	Ebn at Noon	225	248	260	263	267	267	263	258	253	238	213	205	
	Edh at Noon	19	26	34	40	41	41	41	37	31	25	19	17	
All-Sky Solar Radiation	RadAvg	237	454	811	1298	1542	1684	1596	1367	1008	590	275	181	
	RadStd	37	86	124	150	143	156	193	132	124	58	36	25	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page